Geometry-Aware Latent Diffusion Models

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Figure 1: **Decoder-induced latent anisotropy.** Top: reconstruction obtained by decoding the latent code z_0 of a real image. Middle: decoded image after an equal-norm latent perturbation along a low-weight direction. Bottom: decoded image after an equal-norm perturbation along a high-weight direction.

Abstract

Latent Diffusion Models (LDMs) typically assume a Euclidean latent space, despite the decoder inducing a non-uniform pullback metric that leads to anisotropic image-space sensitivity. This mismatch causes the standard denoising objective to weight all latent directions equally.

We introduce **GLDM**, a minimal geometry-aware modification that reweights the DDPM ϵ-prediction loss using a diagonal Hutchinson approximation of the decoder-induced metric evaluated at the clean latent z_0. GLDM requires only two vector–Jacobian products per training step and leaves the diffusion model and sampler unchanged.

On CelebA-64, GLDM improves sample quality across five seeds, reducing FID from 80.3 to 76.7. A random reweighting control does not yield comparable gains, suggesting that the improvement arises from decoder-induced geometry. These results highlight decoder geometry as a useful training signal and motivate geometry-aware diffusion methods.

1 Introduction

Latent Diffusion Models (LDMs) have become a popular approach for efficient high-dimensional image generation by performing diffusion in the latent space of a variational autoencoder (VAE) [Rombach et al., 2022]. By operating in a compressed representation, LDMs dramatically reduce computational cost while retaining high perceptual quality. Despite their success, LDMs inherit a structural assumption from standard diffusion models: that the space in which diffusion occurs is Euclidean.

In the context of VAEs, this assumption is generally violated. The decoder defines a nonlinear mapping from latent space to image space, inducing a pullback Riemannian metric through its Jacobian. As a consequence, equal-magnitude perturbations in different latent directions can lead to highly unequal changes after decoding [Tosi et al., 2014, Arvanitidis et al., 2021]. This anisotropy implies that treating all latent dimensions uniformly misallocates denoising capacity, overemphasizing directions with little image-space effect while underemphasizing sensitive ones. Figure 1 provides a visual illustration of this effect, showing that equal-norm latent perturbations can produce markedly different decoded images depending on direction.

Recent work on diffusion and score-based modeling on Riemannian manifolds demonstrates that respecting underlying geometric structure can substantially alter generative behavior [Bortoli et al., 2022]. However, these approaches typically assume low-dimensional, explicitly parameterized manifolds with access to tractable geometric operators such as exponential maps or metric tensors. Applying full Riemannian diffusion to VAE latent spaces is therefore impractical: the latent dimensionality is high, the geometry is implicit, and computing or storing the full pullback metric is intractable.

In this work, we take a deliberately minimal perspective. Rather than modifying the diffusion dynamics, we ask whether even a coarse, computationally feasible approximation of decoder-induced geometry can improve latent diffusion training. We introduce **GLDM**, a lightweight geometry-aware modification that reweights the DDPM ϵ-prediction loss using a diagonal Hutchinson approximation of the decoder pullback metric evaluated at the clean latent z_0. GLDM leaves the diffusion architecture, noise schedule, optimizer, and sampling procedure unchanged, at the cost of two extra decoder passes per training step.

Through controlled experiments on CelebA-64, we show that GLDM improves sample quality relative to a baseline LDM and outperforms a random reweighting control. These results suggest that decoder-induced geometry provides an actionable training signal even when exploited in a highly restricted form, and motivate further exploration of geometry-aware mechanisms in latent diffusion models.

2 Related Work

Latent diffusion models. Latent Diffusion Models apply diffusion in the latent space of a VAE to reduce the cost of high-dimensional generative modeling [Rombach et al., 2022]. Our work follows this paradigm and focuses on modifying the training objective while keeping architectures and sampling procedures fixed.

Geometry of latent representations. The decoder of a generative model induces a Riemannian geometry on latent space via the pullback metric $G(z) = J_f(z)^\top J_f(z)$ [Tosi et al., 2014]. Prior work has studied geodesics, curvature, and metric-aware regularization in VAEs [Arvanitidis et al., 2021], typically in settings where metric computation is feasible.

Diffusion on manifolds. Several approaches extend score-based generative modeling to Riemannian manifolds [Bortoli et al., 2022]. These methods incorporate metric structure directly into the diffusion dynamics, but generally assume low-dimensional manifolds with tractable geometric operators.

Scalable metric estimation. Exact computation of decoder-induced metrics is infeasible at scale. Stochastic estimators such as Hutchinson's method [Hutchinson, 1989], combined with efficient Jacobian–vector products, provide a practical route to approximate geometric quantities without explicitly forming Jacobians. GLDM adopts this approach to obtain a lightweight diagonal proxy for decoder-induced sensitivity.

3 Method

Our goal is to incorporate decoder-induced geometric information into latent diffusion training without modifying the diffusion architecture or sampling procedure. We achieve this by reweighting the standard denoising objective using a computationally tractable approximation of the decoder pullback metric.

3.1 Latent diffusion objective

Denoising diffusion probabilistic models define a forward noising process

$$q(z_t \mid z_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, z_0,\ (1-\bar{\alpha}_t)I\big)\,, \tag{1}$$

and train a neural network $\epsilon_\theta(z_t, t)$ to predict the noise injected at timestep t [Ho et al., 2020]. In Latent Diffusion Models (LDMs), this process is applied in the latent space of a pretrained variational autoencoder (VAE), allowing diffusion to operate in a lower-dimensional representation [Rombach et al., 2022].

The standard training objective minimizes the mean squared error

$$\mathcal{L}_{\mathrm{DDPM}} = \mathbb{E}_{z_0,\,t,\,\epsilon}\left[\|\epsilon - \epsilon_\theta(z_t, t)\|_2^2\right], \tag{2}$$

which treats all latent dimensions uniformly. This implicitly assumes that equal latent errors correspond to equal image-space distortions, which is an assumption that does not generally hold for learned decoders.

3.2 Decoder-induced geometry

Let $f : Z \to X$ denote the VAE decoder. The decoder induces a Riemannian metric on latent space via the pullback

$$G(z) = J_f(z)^\top J_f(z), \tag{3}$$

where $J_f(z)$ is the Jacobian of f at z. This metric quantifies how infinitesimal latent perturbations are amplified or attenuated when mapped to image space. In practice, the sensitivity encoded by $G(z)$ is highly anisotropic: perturbations of equal magnitude in different latent directions can produce substantially different effects after decoding.

If available, this metric would provide a principled way to prioritize denoising accuracy along latent directions with larger image-space impact. However, forming or storing $G(z)$ is infeasible in high-dimensional latent spaces due to the cost of computing the full decoder Jacobian.

3.3 Geometry-aware loss reweighting

To obtain a scalable proxy for decoder-induced sensitivity, we approximate only the diagonal of the pullback metric using a Hutchinson estimator. For a random probe vector $u \sim \mathcal{N}(0, I)$, we compute

$$\widehat{g}(z) = (J_f(z)^\top u) \odot (J_f(z)^\top u), \tag{4}$$

which provides an unbiased estimate of $\mathrm{diag}(G(z))$ and is averaged over two probes. The required vector–Jacobian product can be computed efficiently using reverse-mode automatic differentiation, without explicitly forming the Jacobian.

We evaluate this estimate at the clean latent z_0 and define normalized geometry weights

$$w(z_0) = \left(\frac{\widehat{g}(z_0)}{\bar{g}}\right)^{1/2}, \qquad \bar{g} = \mathbb{E}_{z_0}\left[\tfrac{1}{d}\textstyle\sum_{i=1}^{d} \widehat{g}_i(z_0)\right], \tag{5}$$

where d is the latent dimension and $\bar{g}$ is approximated by a running batch-wise average. This normalization preserves relative sensitivity across latent dimensions while preventing trivial rescaling of the loss. The square root tempers extreme values of the noisy estimate while keeping their order.

The resulting geometry-aware objective is

$$\mathcal{L}_{\mathrm{geom}} = \mathbb{E}_{z_0,\,t,\,\epsilon}\left[\big\langle w(z_0), (\epsilon - \epsilon_\theta(z_t, t))^2\big\rangle\right]. \tag{6}$$

We refer to this method as **GLDM**. Apart from the loss reweighting, GLDM is identical to the baseline LDM: the diffusion architecture, noise schedule, optimizer, and sampling procedure are unchanged.

4 Implementation Details

Variational autoencoder. All experiments use a convolutional variational autoencoder trained on CelebA images resized to 64×64. The VAE latent dimensionality is fixed across experiments. During diffusion training, all VAE parameters are frozen, ensuring that observed differences arise solely from modifications to the diffusion objective rather than changes in the latent representation.

Latent noise-prediction network. The noise-prediction model ϵ_θ is implemented as a lightweight UNet operating directly on latent vectors. Architectural details, initialization, and parameter counts are identical for all methods, including the baseline LDM, GLDM, and the random-weight control.

Diffusion process. We employ a standard DDPM forward process with a linear β-schedule over $T = 1000$ diffusion steps and use the ϵ-prediction parameterization throughout. The diffusion noise schedule and timestep sampling strategy are shared across all experiments.

Training procedure. Models are trained for two epochs using the Adam optimizer with learning rate 5×10^{-4} and batch size 64 on a single GPU, with five random seeds per method. The short schedule limits absolute sample quality but allows for a controlled comparison of relative performance across methods.

Random-weight control. To distinguish geometry-specific effects from generic loss reweighting, we include a control variant in which the geometry-derived weights $w(z_0)$ are replaced with positive random weights. These random weights are normalized using the same procedure as in GLDM, isolating the contribution of decoder-induced geometry.

Evaluation. Sample quality is evaluated using Fréchet Inception Distance (FID) computed between 5,000 generated samples and the first 5,000 CelebA-64 test images. All models are evaluated using identical sampling procedures and evaluation code paths.

5 Results

We compare GLDM against a baseline Latent Diffusion Model (LDM) under identical training, architecture, and sampling conditions. Given the scope and computational constraints of our setup, we focus on sample quality as measured by Fréchet Inception Distance (FID), which serves as our primary evaluation metric. All results are reported on the CelebA-64 test set.

5.1 Sample quality

Table 1 reports FID scores computed using $5{,}000$ generated samples for each model, averaged over five seeds. Under identical training procedures and diffusion settings, GLDM improves sample quality relative to the baseline LDM in every seed, reducing FID from 80.3 to 76.7.

Table 1: Sample quality on CelebA-64 measured by FID using $5{,}000$ generated samples (mean $\pm$ standard deviation over five seeds).

Model	FID $\downarrow$
LDM (baseline)	80.28 ± 1.58
Random-weighted loss (control)	80.05 ± 1.33
GLDM (ours)	$\mathbf{76.70 \pm 0.98}$

The random-weight control yields almost no improvement over the baseline (80.05 versus 80.28), indicating that reweighting the denoising errors alone does not explain the gain.

Overall, GLDM attains the lowest FID among the evaluated methods in every seed. Since random reweighting has little effect, the improvement observed for GLDM suggests that incorporating structure derived from the decoder-induced geometry provides a more informative training signal than unstructured or random weighting in this setting. In a longer run (20 epochs, three seeds), the

baselines need about 20 epochs to approach the FID that GLDM reaches after two, so its advantage is faster training rather than a lower final FID.

6 Limitations and Future Work

While GLDM demonstrates that decoder-induced geometry can improve latent diffusion training with minimal overhead, several limitations remain. First, geometric information is incorporated only through the training objective; the reverse diffusion dynamics themselves remain Euclidean. As a result, sampling does not explicitly follow the geometry induced by the decoder, and generated trajectories do not correspond to geodesics in latent space.

Second, GLDM relies on a coarse diagonal approximation of the pullback metric obtained via a Hutchinson estimator. This approximation ignores correlations between latent dimensions and introduces stochastic noise into the weight estimates. Although our results indicate that this signal is sufficient to yield measurable improvements, more accurate or structured metric approximations may further enhance performance at increased computational cost.

Third, our evaluation focuses on Fréchet Inception Distance as a primary measure of sample quality. While FID is widely used and appropriate for unconditional image generation, incorporating complementary perceptual or task-specific metrics would provide a more comprehensive assessment in future work.

Finally, our experiments are limited to unconditional image generation with a fixed, compact VAE trained under constrained compute. Extending geometry-aware diffusion to larger decoders, conditional or text-guided settings, and alternative latent representations represents a promising direction for future work. In particular, incorporating metric information directly into the reverse diffusion process, such as through preconditioned updates or geometry-aware noise scaling, could more fully exploit decoder-induced geometry.

7 Conclusion

We investigated whether decoder-induced geometry can improve latent diffusion training in high-dimensional VAE latent spaces. By reweighting the DDPM ϵ-prediction loss using a diagonal Hutchinson approximation of the pullback metric evaluated at the clean latent z_0, GLDM introduces geometry awareness with modest computational overhead and no changes to the diffusion architecture or sampler.

Despite its simplicity, GLDM improves sample quality on CelebA-64 under controlled experimental conditions. These findings suggest that decoder geometry provides an actionable signal for guiding latent diffusion training, even when leveraged in a highly restricted form, and motivate further exploration of geometry-aware mechanisms in latent diffusion models.

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